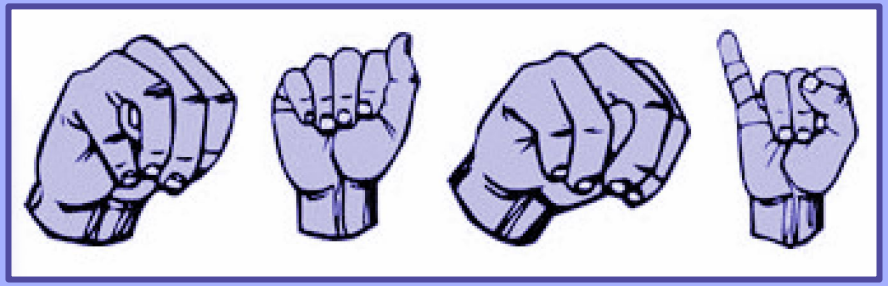
Project MANI



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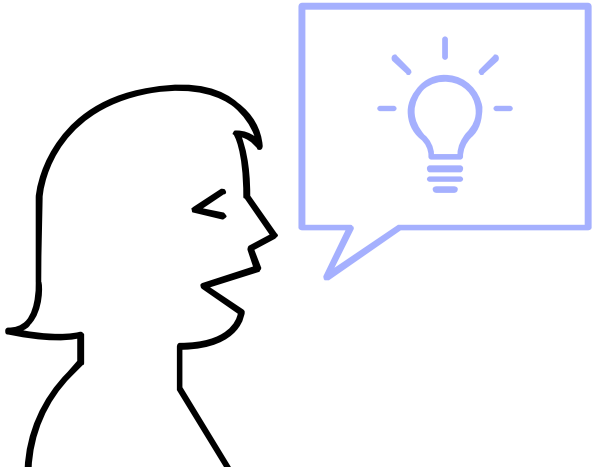
The Problem



- Home Virtual Assistants (VA's) such as Alexa, or Google Home are incredibly convenient, but the Deaf and Hard-of-Hearing do not benefit from this technology.

- Over 1,000,000 Americans are considered "functionally deaf" with over 500,000 using American Sign Language (ASL) as their primary language. As such, they can't benefit from the convenience of this hearing-based technology.

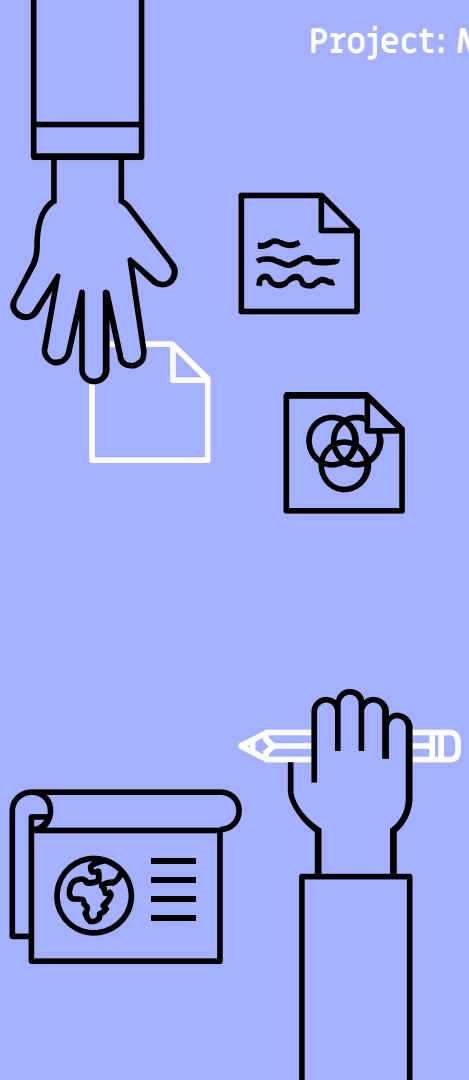
- Some current VAs offer a smart-speaker with a touchscreen, but this still requires users to physically interact with the device, negating the convenience

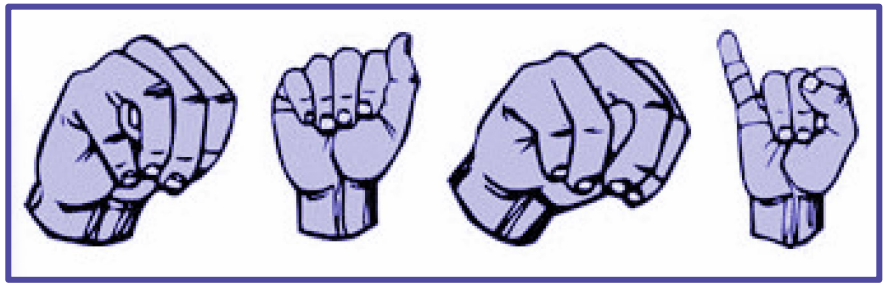
Idea

Create a Home Virtual Assistant for the Deaf which gives users the **features and convenience** of a typical home VA, through **visual**, instead of audible, communication.

Users

The Deaf community (extreme hearing loss such that ASL is their **primary language**), and their families.





Presenting: MANI

Instead operating with a microphone and speaker, Mani would operate with a **display** and a rotating **camera**. Users would raise their hand, sign to the camera, have their requests processed, and the results displayed visually.

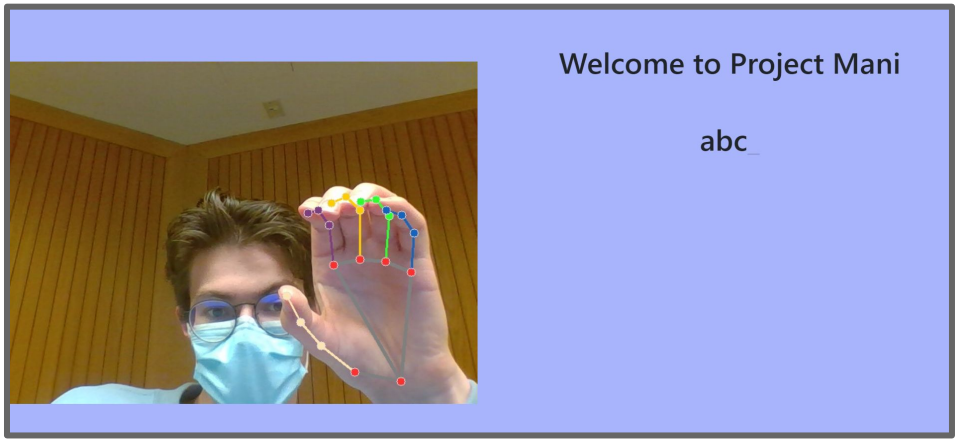
For our MVP, we will focus on one-handed **fingerspelling**

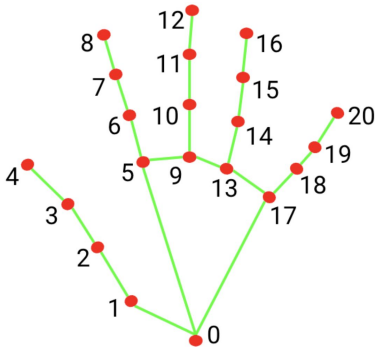
In this way, Deaf users would be able to utilize the same **functionality and convenience** of VAs that hearing people already benefit from.

Current Functionality

Currently, MANI can do the following:

- Display a live camera feed to the user.
- Recognise ASL fingerspelling with 86% accuracy on static images, with higher accuracy on live video.
- Output the fingerspelled letters to the user on the interface in real time.
- Connect to the Google Assistant to process the results of the command
  - Including fully spelled commands and hard-coded short-cuts.
- Delete a letter in the command-line if the letter 'x' is spelled ("backspace function").





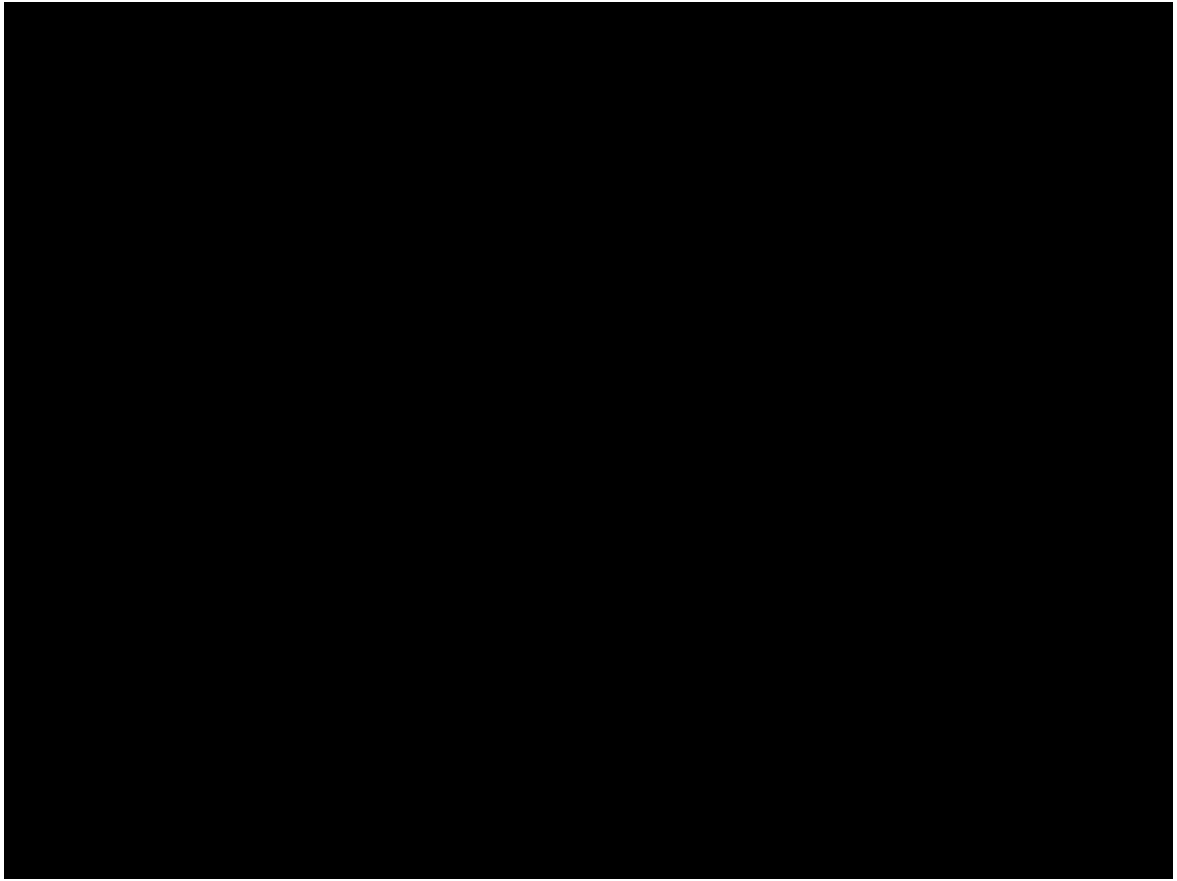
Google Assistant

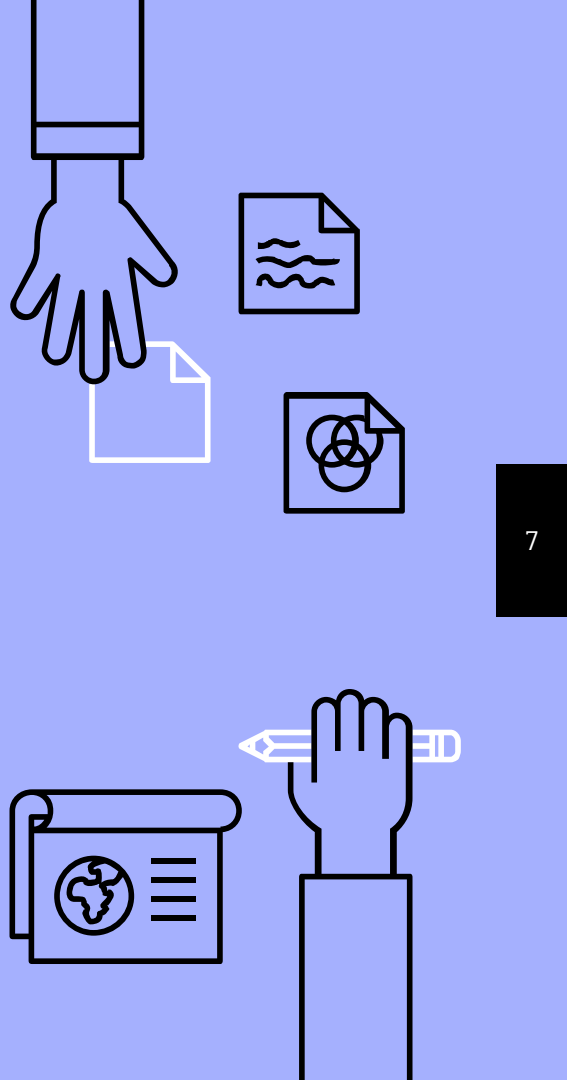
Fingerspelling Shortcuts

Added the following shortcuts so users can input common commands without fully fingerspelling them:

- **"wh"**: "What is the weather in Hanover, New Hampshire?"
- **"poki"**: "How many pounds in a kilogram?"
- **"hol"**: "When is the next federal holiday?"
- **"cd"**: "What is the current date?"
- **"gl"**: "Turn on the garage lights"
- **"al"**: "Arm the house alarm"
- **"am"**: "Add milk to my shopping list"
- **"tse"**: "Set the thermostat to 68 degrees"
- **"asam"**: "Set an alarm for 7 AM"
- **"bc"**: "What is the news from the BBC?"
- **"dan"**: "What is the definition of anthropomorphize?"
- **"hlw"**: "How large is a blue whale?"
- **"cal"**: "What's on on my calendar"
- **"cnn"**: "What's the news from CNN?"
- **"gs"**: "What is the nearest grocery store?"

Demo





Goals for Spring Term 2022

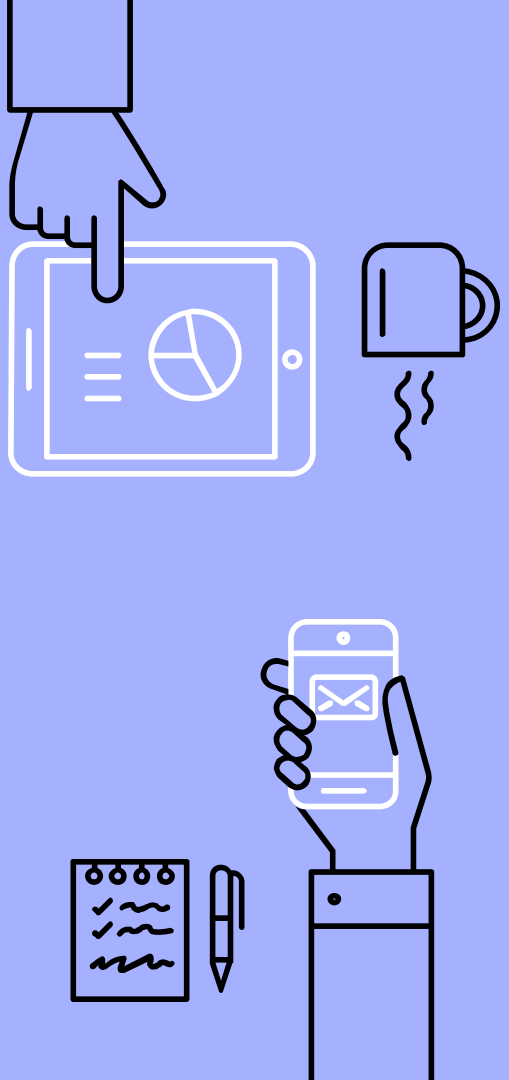
Improvements to the Hardware

- Run MANI on dedicated hardware
- Add rotating camera and movement tracking for full-room capability

Improvements to the Display

- Help Screen
- Custom Shortcuts Screen
- Prettier responses from Google Assistant

Google Assistant

Improvements for usability

- Get feedback from members of the Deaf community
- Improve model performance
- Clearer start and stop user flow
- Make shortcuts more flexible